

Bulk-color v0.3 — team integration: Elie's $[p_+, p_+] = 0$ obstruction (Toy 3612) + Grace's G10 four-family map. My v0.2 SO(3) sub-vector candidate maps to Family (4) counting-not-symmetry — NOT requiring a Lie subgroup acting on p_+ — so it's CONSISTENT with the Hermitian-abelian obstruction. The candidate sharpens, NOT contradicts.

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Bulk-color v0.3 — team integration

0. Team broadcasts since v0.2

While I was working morning items, Elie and Grace pushed bulk-color complements:

- Elie Toy 3612 (A1/P1.1): SO(5,2) Cartan decomposition. $\dim p = 10$; $p = V_{-5} \otimes (V_{+1} \oplus V_{-1})$ under $K = SO(5) \times SO(2)$; $p_+ = V_{-5} \otimes V_{+1}$ (holomorphic tangent). $[p_+, p_+] = 0$ — HERMITIAN ABELIAN. This is a STRUCTURAL OBSTRUCTION to embedding any non-abelian Lie subgroup acting on p_+ alone.
- Grace G10 v0.1: catalog scaffold organizing FOUR candidate mechanism families for bulk-color SU(3):
 - (1) K-action on p (non-abelian Lie subgroup) — OBSTRUCTED by Elie's finding.
 - (2) Bulk K-type tower (Wallach spacing) — my Direction (A).
 - (3) Kernel modular structure (Bergman $Z/3$ residue) — my Direction (C).
 - (4) Emergence as counting (counting-not-symmetry) — Casey's Track P direction.

1. Where my v0.2 SO(3) sub-vector candidate fits

My v0.2 proposed: COLOR = SO(3) $\subset$ SO(5) sub-vector 3-fold in the bulk holomorphic tangent $T^{(1,0)} = 5_{+1}$. The 5-vector of SO(5) decomposes under SO(3) $\times$ SO(2) as $(3,0) + (1,\pm 1)$ — a 3-dim spatial vector subspace.

Critical clarification (in light of Elie's finding): this is FAMILY (4) counting-not-symmetry, NOT Family (1).

- The “3” comes from the 3 SPATIAL DIRECTIONS in the SO(3)-vector subspace — a labeling/counting structure on the 5-dim tangent.
- It does NOT require SO(3) to act on p_+ as a non-abelian Lie subgroup (which would conflict with $[p_+, p_+] = 0$).
- SO(3) $\subset$ SO(5) is a subgroup of K (not of p_+); it acts on p_+ via the V_{-5} representation, which IS abelian as a tangent-vector action (because p_+ itself is abelian as a Lie algebra). The SO(3) subgroup picks out 3 directions; this picking is COMPATIBLE with the abelian p_+ structure.

So Elie's $[p_+, p_+] = 0$ finding is consistent with my v0.2 candidate, and clarifies its TYPE: it's a counting-not-symmetry mechanism, not a gauge-Lie mechanism. This matches Casey's Track P / Family (4) framing exactly.

2. Updated candidate map (v0.3)

Family	Status	My contribution
(1) Non-abelian Lie on p_+	RULED OUT (Elie's $[p_+, p_+] = 0$)	—
(2) Bulk K-type tower Wallach spacing	Open	P4.3 framework set up; bulk tower $V_{(k,0)}$ Casimirs 0,4,10,18,... computed
(3) Bergman $Z/3$ residue	Likely dead end	no clean $Z/3$ found in D_{IV^5} 's modular structure (v0.1 mapping)
(4) Counting-not-symmetry	My v0.2 candidate ($SO(3)$ sub-vector 3-fold)	avored direction; consistent with $[p_+, p_+] = 0$ obstruction

Favored direction (Lyra v0.3, consistent with team findings): Family (4) counting-not-symmetry, instantiated as $SO(3) \subset SO(5)$ sub-vector 3-fold in the bulk tangent. The 3 directions in the $SO(3)$ -vector subspace are the color labels; the substrate's "color" is this 3-direction counting structure, not a non-abelian gauge symmetry acting on p_+ .

3. The remaining open work ($SU(3)$ emergent gauge phenomenology)

Even with Family (4) clarified, the $SU(3)$ emergence is the open burden: how does a 3-direction counting ($SO(3)$ sub-vector) produce the OBSERVED 8-fold gluon phenomenology of low-energy QCD?

Honest assessment after morning work + integration:

- 3-fold direction labeling is in hand ($SO(3)$ sub-vector, Family 4).
- 8 gluons emergence is NOT in hand. The natural candidates ($3 \times 3 = 9 - 1 = 8$ from removing the trace $U(1)$) require connecting the 3-direction labeling to the gauge-field structure of QCD — which is multi-week research.
- One direction worth exploring: $SU(3)_{\text{emergent}} = SO(3) \otimes SO(3)$ symmetric-traceless = 5-dim spin-2 OR (3,3)-bifundamental — none cleanly gives 8.
- Another direction: the 8 gluons come from the (color $\times$ generation) cross-cutting tensor ($3 \times 3 = 9 - 1 = 8$ after removing the $U(1)$ trace). This would tie color and generations more tightly than the "two independent 3-folds" picture; would need to revisit Keeper's burden.

Either way, the 8-gluon emergence is the next deep target, and it's multi-week.

4. Routing for the team

- → Elie: your $[p_+, p_+] = 0$ finding cleanly rules out Family (1) candidates (non-abelian gauge in p_+). My v0.2 $SO(3)$ is Family (4), consistent with the obstruction. Next: if you can compute the K-action of $SO(3) \subset SO(5)$ on higher bulk K-types (Sym^k of 5-vector), it tests whether the $SO(3)$ substructure extends naturally up the tower.
- → Grace: my v0.2 is your Family (4); G10's catalog scan for $SU(3)$ -emergent counting structures has direct relevance. Your G10 + Mendelev pair analysis is the right framework for cataloging candidates.
- → Keeper: bulk-color disposition now: Family (1) RULED OUT; Family (4) FAVORED (Lyra v0.2 + Grace G10 + Casey Track P aligned); Families (2)+(3) still open as alternates. The 8-gluon emergence is the open burden.
- → me: integration done v0.3; the $SU(3)$ gauge-emergence-from-counting derivation is the remaining multi-week item. Continuing to other items in the afternoon.

5. Honest scope + tier

RIGOROUS (now multi-source verified): - $SU(3) = A_2 \not\cong SO(5) = B_2$ (rank-2 algebras differ); $SU(3)$ not in K. - $[p_+, p_+] = 0$ (Elie Toy 3612) — Hermitian abelian; rules out Family (1). - $SO(3) \subset SO(5)$ subgroup with branching $5 = (3,0) + (1,\pm 1)$ under $SO(3) \times SO(2)$. - p_+ K-decomposition: $p_+ = V_{-5} \otimes V_{+1}$.

CANDIDATE (v0.3, sharpened): COLOR = Family (4) counting-not-symmetry, instantiated as SO(3) sub-vector 3-fold in the bulk tangent. Consistent with all team findings.

OPEN burden (multi-week): derive 8-gluon SU(3) gauge phenomenology from 3-direction substrate counting.

Cal #27 / honesty: I'm NOT claiming SU(3) emergence is derived. I'm aligning my v0.2 candidate with the team's findings (Elie's obstruction + Grace's family map); the SO(3) sub-vector candidate is now in Family (4) counting-not-symmetry, consistent with the constraints, with the 8-gluon emergence as the named open burden. Discipline: candidate sharpens, the harder part stays open.

— Lyra, bulk-color v0.3 team integration. Elie's $[p_+, p_+] = 0$ (Toy 3612) RULES OUT Family (1) candidates (non-abelian Lie subgroup acting on p_+) per Grace's G10 four-family map. My v0.2 SO(3) sub-vector candidate is Family (4) counting-not-symmetry — 3 SPATIAL DIRECTIONS in the SO(3)-vector subspace as color labels, NOT a Lie subgroup acting non-abelianly on p_+ . Consistent with the obstruction; candidate sharpens, NOT contradicts. Favored direction: Family (4) counting-not-symmetry. Open burden: derive 8-gluon SU(3) gauge phenomenology from 3-direction counting (multi-week).